Eric Xiao, EIT

ericxiao.me@gmail.com | linkedin.com/in/eric-xiaoy | (917) 562 - 8538 | ericxiao.me

Experience

Robotics & Controls Engineer — January 2026 – Present
Kathedra – Research & Development (R&D) — *Hickory, NC*

- Architected full-stack robotic control systems for a six-figure proof-of-concept workcell, validating a ∼30% cycle time reduction against manual baselines.
- Developed the hardware and software integration for a custom 7th-axis linear rail, designing custom PCBs and motor driver interfaces to enable dynamic motion planning and extended reach.
- Engineered a 3D vision pipeline on NVIDIA Jetson using PCA, RANSAC, and ICP to extract object surface normals and generate real-time, vision-informed trajectories.
- Filed patent-pending designs for a custom end-effector and application toolhead, driving end-to-end SolidWorks CAD development, CNC machining, mechatronics integration and 3D printed rapid prototyping to deliver a proof-of-concept.

Experimental Systems Research Assistant — September 2025 – December 2025
Stony Brook University – Interacting Robotic Systems Laboratory — *Stony Brook, NY*

- Developed data acquisition pipelines to drive a U-Net ML model, fusing FEA-simulated deformation data with real-world visual sensor streams for sim-to-real validation and real-time force estimation.
- Engineered custom physical validation setups executing repeatable motion to benchmark model accuracy within 5%.

Mechanical Engineer Intern — June 2025 – December 2025
Brookhaven National Laboratory – National Synchrotron Light Source II — *Upton, NY*

- Architected electro-mechanical automated liquid dispensing systems, developing custom RS485 driver stacks to achieve $5\mu L \pm 0.5\mu L$ accuracy; deployed in compliance with Department of Energy beamline safety standards.
- Engineered a high-speed data streaming stack to perform teleoperation, mapping a Standard Open arm to a UR5e cobot; facilitated high-fidelity data collection to train Action Chunking Transformer policies for imitation learning.

Embedded Systems and Controls Engineer — June 2024 – May 2025
Stony Brook Rocket Team – NASA Student Launch Initiative — *Stony Brook, NY*

- Engineered a multi-threaded telemetry system in C++ for a resource-limited payload, encoding flight data into APRS packets and validating transmission with Direwolf to ensure reliable real-time data downlink.
- Led the integration of avionics, power distribution, and controls through NASA safety and design reviews.

Projects

Dynamic Trajectory Planning & Control | *Python, Control Theory, Kinematics* — November 2025 – December 2025

- Implemented forward and inverse kinematics for an AUBO i5 robot, developed task-space motion planning algorithms and Jacobian-based velocity control for constrained end-effector trajectories.
- Performed Monte Carlo simulations to estimate the robot workspace and analyze kinematic singularities.
- Designed and simulated time-parameterized joint and Cartesian trajectories to evaluate motion feasibility and stability.

Autonomous UAV Control & Navigation System | *Sensor Fusion, Control Theory* — September 2025 – November 2025

- Engineered a lightweight data-fusion pipeline for a resource-constrained platform, combining IMU data with spatial sensors for real-time environmental mapping using sparse sensing SLAM.
- Implemented cascaded controllers to process minimum snap trajectory setpoints for smooth movement.

Skills

Certificates & Awards: NYS Engineer In Training, Student Choice Award 2025 – Stony Brook, Hack @ CEWIT 2023: Winner
Programming: Python, C++, Java, SQL, MATLAB, TCP/IP, UART, I2C, EtherCAT, CANBus
Software & Engineering Tools: SolidWorks, NX, AutoCAD, Cura, LabVIEW, Epilog Laser, Bash, Git, Linux, Docker, DFM, FEA
Robotics & Simulation: ROS2, Motion Planning, URDF, RViz, ros2_control, Flight Dynamics, UAV Control, SLAM
Controls: Cascaded Control, Sensor Fusion, Control Theory, Kalman Filters, System Identification
Data & ML: Pandas, NumPy, Matplotlib, OpenCV, scikit-learn, PyTorch, PCL, CUDA

Education

Stony Brook University — Stony Brook, NY
Master of Science in Mechanical Engineering — *September 2024 – December 2025*
Bachelor of Science in Mechanical Engineering — *September 2020 – May 2024*